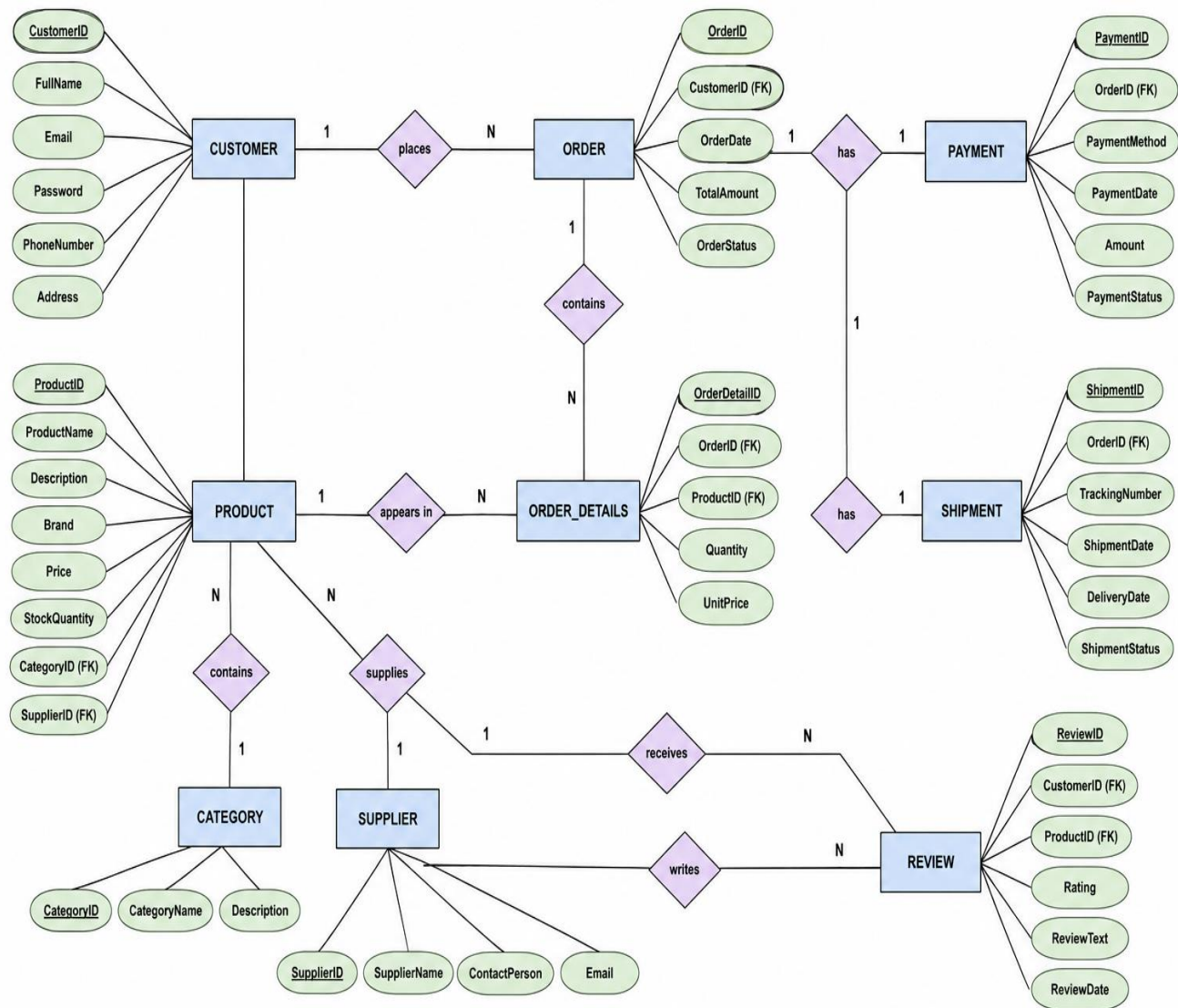


Entity Relationship (ER) Diagram

Project Title: Online Shopping Database Management System (Implementation: SmartShop)



Explanation of the ER Diagram

1. Entities

- Entities represent real-world objects whose data is stored in the database.
- The entities used in this ER diagram are:
 - Customer
 - Product
 - Category
 - Supplier
 - Order
 - Order_Details
 - Payment
 - Shipment
 - Review

2. Attributes

- Attributes are the properties or characteristics of an entity.
- They store information related to each entity.
- Examples:
 - Customer: CustomerID, FullName, Email, PhoneNumber
 - Product: ProductID, ProductName, Price, StockQuantity

3. Relationships

- Relationships show how different entities are connected.
- Examples:
 - Customer places Order
 - Supplier supplies Product
 - Order contains Order_Details

- Customer writes Review
- Product receives Review

4. Primary Key (PK)

- A Primary Key uniquely identifies each record in an entity.
- It does not allow duplicate or NULL values.
- Examples:
 - CustomerID
 - ProductID
 - OrderID
 - PaymentID
 - ShipmentID

5. Foreign Key (FK)

- A Foreign Key links one entity to another by referring to its Primary Key.
- It helps maintain relationships between tables.
- Examples:
 - CustomerID in Order
 - CategoryID in Product
 - SupplierID in Product
 - OrderID in Payment and Shipment

6. Cardinality

- Cardinality defines how many records of one entity are associated with another entity.
- The cardinalities used in this ER diagram are:
 - Customer (1) → Order (N)
 - Category (1) → Product (N)
 - Supplier (1) → Product (N)

- Customer (1) → Review (N)
- Product (1) → Review (N)
- Order (1) → Payment (1)
- Order (1) → Shipment (1)
- Order (1) → Order_Details (N)
- Product (1) → Order_Details (N)